# 4G OBD Product Specification

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# 1. Product photo and Description


CC550 4G intelligent Device meets OBDII diagnostic standard, communicates with Vehicle ECU computer, and has remote diagnosis function; Integrated with advanced wireless all network data (2G / 3G / 4G) communication module, with 4G vehicle WiFi hotspot sharing; High precision GPS module can provide accurate positioning; High performance three-axis gravity sensor can obtain driving behavior analysis and collision warning; It is free of installation and plug and play. It is connected to the vehicle diagnosis interface to obtain the position information, vehicle diagnosis data and fault information in real time. It uploads the vehicle operation status, vehicle condition data, vehicle fuel consumption data, driver control data, etc. to the service platform. The service platform analyzes these data, statistics the vehicle trajectory, driving behavior, fuel consumption analysis, anti-theft tracking, etc.

# 2. Product characteristics

● Easy Installation, which does not need to damage the line of the original vehicle and cause damage to the original vehicle.

● it supports a wide range of Vehicle models to collect data, and can adapt to more than 95% of the models on the market.

● the sleep and wake-up mechanism of low power consumption can greatly reduce the impact to original vehicle battery.

● remote monitoring of driving behavior, real-time vehicle speed, engine load and Built in 3-axis accelerometer.

● real time monitoring of vehicle fault code, vehicle battery, remaining fuel volume and vehicle safety status.

● obtain the information of fuel consumption and mileage from the automobile BUS, with an accuracy of 99%.

● vehicle fault alarm: real-time monitoring of engine sensors and actuators, real-time alarm.

● low voltage alarm of vehicle battery. If the Low battery voltage is detected, the alarm will be sent to the owner.

● Main power disconnect alarm: device will send alarm to the vehicle alarm when it pull out of OBD interface.
● vehicle track tracking, query the current position information of the vehicle at any time and place, and play back the historical running track.
● butterfly algorithm is adopted to optimize the GPS trajectory, so that the GPS trajectory will not drift and the positioning is accurate.
● supplementary report of blind area data: the train engine will store the information, and then transmit the data to the report after there is a signal.

## 3. Product Functions

| | | |
|---|---|---|
| OBD | Vehicle data flow | Regularly report the vehicle key data flow. |
| | Read vehicle fault | The device identifies the vehicle fault information and reports it to the platform when the vehicle fault status changes. |
| | Clear vehicle fault code | Platform send command to Clear the vehicle fault code |
| | Ignition on/off | Accurately judge the ignition on/off, and enter sleep mode after ignition off to reduce power consumption. |
| | Mileage and fuel consumption statistics | Accurate calculation and statistics of vehicle mileage and fuel consumption by reading OBD data. |
| | Travel Report | After the end of a trip, will report the vehicle information of this trip. |
| | Vehicle physical examination | It can realize professional fault diagnosis of vehicles and provide a complete vehicle condition assessment. |
| GPS positioning | GPS positioning | Precise GPS positioning |
| | Data | Longitude and latitude, time, number of satellites, speed, direction, acc status, battery voltage etc |
| | LBS | Including network base station lac, cell ID information and operator code. |
| | Intelligent track | The terminal automatically judges the track route and achieve the butterfly effect of the track. |
| | Time interval report | The monitoring platform sets the mode and frequency parameters of sending the location information back from the device |
| Network communicati | Network | LTE-FDD B1/B2/B3/B4/B5/B7/B8/B12*/B13*/ |

| | | |
|---|---|---|
| on | | B18*/B19*/B20/B25*/B26*/B28/B66/<br>LTE-TDD B34/B38/B39/B40/B41<br>GSM/GPRS/EDGE<br>850/900/1800/1900 MHz<br>*optional |
| | WiFi hotspot sharing | WiFi hotspot Internet access for passengers in the vehicle (optional) |
| | Reconnect network | If the network is disconnected, it can be automatically reconnected. |
| | Protocol | TCP protocol is adopted for data information and platform interaction |
| | Parameter setting | Set the IP address, port number and APN of the device directly through the platform. |
| Driving behavior | Driving behavior | Including the start and end time of the driving cycle, the total mileage of the driving cycle, the average speed, the maximum speed, the overspeed duration, the over speed times, the idle duration, the sharp acceleration, the sharp deceleration and the sharp turn information. |
| | Events in driving | Will separate report the event During the driving process of the vehicle, rapid acceleration, rapid deceleration and rapid turning events, fatigue driving etc. |
| Reminder alarm | Vehicle fault alarm | Through OBD diagnosis, the vehicle has fault code alarm. |
| | ignition on alarm | When ignition on, device will send ignition on alarm to platform, including ignition time and GPS information. |
| | Ignition off alarm | When ignition off, device will send ignition off alarm to platform, including ignition time and GPS information. |
| | Low voltage alarm | If the vehicle battery voltage is lower than the settable value, a low voltage alarm will be reported, including the alarm time and GPS information. |
| | long Idle speed alarm | If the idle time of the vehicle exceeds the settable time, an alarm of too long idle time will be reported, including the alarm time and GPS information. |
| | Water temperature alarm | For vehicles that support OBD water temperature data, if the water temperature exceeds the settable temperature threshold, a water temperature alarm will be reported, including the alarm time, GPS information and the current water temperature value. |
| | Over speed alarm | If the vehicle running speed exceeds the settable speed threshold, an overspeed alarm will be reported, including the alarm time and GPS information. |
| | Collision alarm | When the vehicle is running, the generated acceleration is greater than the settable threshold value. After filtering the |

| | | |
|---|---|---|
| | | front and rear vehicle speeds, it is defined as a serious collision scenario, and a collision alarm is reported, including collision time, GPS information and collision acceleration |
| | Power on alarm | When plug in the device on OBD port, device will send power on alarm to platform |
| | Main power disconnect alarm | When the device is pulled out, the main power disconnect alarm is reported (which is available when device has built in battery). |
| Remote control management function | Remote upgrade | The device firmware can be upgraded remotely through the network and through the FTP server. |
| | Remote Reboot | Send command to restart the device through data channel. |
| | Remote query | The platform remotely inquires device information through network, such as vehicle type, GPRS communication parameters, heartbeat parameters, etc. |
| | Remote settings | The platform remotely sets the terminal information, vehicle type, GPRS communication parameters, heartbeat parameters, etc. through the mobile network. |
| Platform function | Docking protocol | Dixin 18GPS platform cywobd air protocol<br>Jt808 China ministerial standard platform with OBD extended data<br>New energy vehicle GB / T 32960 protocol (customized) |

## 4. Application fields

Individual user
Vehicle selling company
Operators
Insurance company
Vehicle rent company
Transportation company
Taxi Company

## 5. Technical Information

| | | |
|---|---|---|
| Mechanical parameters | size | 67mm（length） x 53mm（width） x 27mm（height） |
| | weight | 50g |
| Interface | | OBD interface: Standard 16pin OBD II<br>SIM card interface: Micro SIM, push push type<br>Debugging setting port: type-C USB |

| | | |
|---|---|---|
| Storage | | 4MB flash, up to 20000 pieces of GPS data can be stored |
| Diagnostic protocol | | ISO 9141-2<br>ISO 14230-4<br>ISO 15765-4<br>SAE J1939（Optional）<br>SAE J1587/J1708（optional） |
| Power | Working voltage | 9-36VDC With anti reverse connection function and self recovery fuse |
| | Average operating current： | <150mA |
| | Maximum operating current： | <200mA |
| | Sleep mode current： | <3mA |
| | Built in battery | 3.7V/90mAH（optional） |
| Triaxial acceleration sensor | | +/-16g，Driving behavior detection |
| Positioning mode | | Support AGPS assisted positioning |
| GPS | | Receiving channel: 48<br>Receiving sensitivity: - 163dbm<br>Positioning accuracy: 5m CEP<br>Cold start: < 60s half sky state<br>Hot start: < 1s |
| Data transmission mode | | LTE/ HSPA/ GPRS/ SMS |
| 4G Frequency band | Global | LTE-FDD<br>B1/B2/B3/B4/B5/B7/B8/B12*/B13*/<br>B18*/B19*/B20/B25*/B26*/B28/B66/<br>LTE-TDD B34/B38/B39/B40/B41<br>GSM/GPRS/EDGE<br>850/900/1800/1900 MHz |
| | Asian | LTE-FDD B1/B3/B5/B8<br>LTE-TDD B34/B38/B39/B40/B41<br>GSM/GPRS/EDGE<br>900/1800 MHz |
| | Europe | LTE-FDD B1/B3/B5/B7/B8/B20<br>GSM/GPRS/EDGE<br>900/1800 MHz |
| | Australia | LTE-FDD<br>B1/B2/B3/B4/B5/B7/B8/B28/B66<br>GSM/GPRS/EDGE |

| | | |
|---|---|---|
| | | 850/900/1800/1900 MHz |
| WiFi sharing | | IEEE 802.11a/b/g/n/ac wireless transmission protocol<br>Generate WiFi hotspot and share network under 4G network |
| Indicate light | | OBD/4G/GPS indicate light |
| antenna | 4G antenna | Built in PIFA |
| | WiFi antenna | Built in PIFA |
| | GPS antenna | Built in 25*25*4mm ceramics antenna |
| Extensible function | | Bluetooth ble4.0<br>RF wireless relay, RF wireless vehicle key |
| Environmental parameters | Working temperature | -20℃ ~ +70℃ |
| | Storage temperature | -40℃ ~ +85℃ |
| | Humidity | 5%～95%（No frost） |
| | Muggy | GBT2423 JTT794 GB/T28046 GB/T28046.2 |
| | electromagnetic compatibility | Meet the requirements of GB/T17619、GB/T18655、GB/T19951、GB/T21437 |

## 6. Meaning of indicate light

| Color of indicate light | Flashing status | Meaning |
|---|---|---|
| Blue（OBD light） | Flashing | Attempting to establish a communication connection with the vehicle |
| | Keep on | Reading vehicle operation data |
| red（network light） | flashing | Attempting to establish a communication connection with the server |
| | Keep on | Transferring data to server |
| green（GPS light） | Flashing | Try to Locating |
| | keep on | Successfully located |

## 7. SMS query setting command

| | |
|---|---|
| AT+HOSTS? | Query IP and port |
| AT+HOSTS=8.8.8.8,8888 | Set IP and port |

| | |
|---|---|
| AT+APN? | Query APN |
| AT+APN=CMNET,user,pass | Set APN, user name and password can be omitted |
| AT+VERSION? | Query Firmware |
| AT+MAP | Query location |
| AT+RESET | Reset device |

## 8. Precautions for use and installation


Common OBD interface position of Virous Vehicle Models

A area: Most models of GM, Volkswagen, Ford, Toyota, Hyundai, Citroen, BMW and other brands

B area: Honda, and Lexus

C area: Dongfeng Citroen, Dongfeng Peugeot and other few brand models

D area: Dongfeng Citroen and other few brand models

E area: Other few brand models

# OBD Interface Diagram